Reciprocal pooling across signed bases for Diophantine consistency

AI-assisted research prepared for Vladimir Reshetnikov
A contribution for the ProveIt repository

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Abstract

We give a uniform arithmetic compression theorem for finitely many Pell-certified power requests with different, signed, or zero bases. A single polynomial congruence witness replaces the separate witnesses for all power blocks. The principal new point is a second integrality argument: it rules out cancellation between positive and negative reciprocal moduli without introducing orientation factors. Applied to an explicit, calibrated numerical proof-calculus equation in the ProveIt repository, it removes nine coordinates. Subsequent exact monic eliminations give the fixed integer polynomials listed below. For every delivered equation, ordinary mathematics formalizable in ZFC, composed with the explicitly retained source calibration, proves that absence of an integer zero is equivalent to syntactic Con(ZFC). The complete finite polynomial definitions, source residuals, elimination tables, executable circuits, and independent finite checkers are supplied. We do not claim a globally simplest equation, a consistency proof, or a new end-to-end proof-assistant certificate.

1 Results and the meaning of simplicity

Our principal recommendation is the fixed polynomial in **112 unrestricted integers of exact degree 24**. A lower-degree companion uses **119 integers at degree 16**; the smallest variable count in this release is **102 integers at degree 64**. Each has the required ZFC equivalence, with the dependency boundary described below.

Integer variables	Exact degree	Residuals	Height bound	Binary operations
143	8	80	$< 2^{43}$	2590
119	16	56	$< 2^{79}$	2519
112	24	49	$< 2^{79}$	2501
107	32	44	$< 2^{113}$	2482
102	64	39	$< 2^{214}$	2437

The 143-variable octic is a useful companion obtained with five singleton blocks and ordinary elimination; it does not need the new cross-base fusion theorem. The other four rows use that theorem. Operation counts describe the delivered literal evaluators and are not claimed optimal.

For comparison, package 152 gives 153 integers at degree 8, 125 at degree 16, 117 at degree 32, and 111 at degree 64 [4]. The corresponding supplied coefficient-height bounds are $2^{47}, 2^{80}, 2^{144}, 2^{173}$. Our results lower the variable counts at those degrees; our degree-64 height bound 2^{214} is larger than the older bound 2^{173}. Thus even this direct-family comparison has a coefficient tradeoff.

A fair local control isolates the new arithmetic gain. Apply exactly the principal form's 95 core eliminations to the original source and also eliminate its three modulus coordinates and two already monic K coordinates. This gives 116 integers, 53 residuals, and exact degree 24 without cross-base fusion. The new construction has 112 integers, 49 residuals, and the same degree. The four-coordinate saving in this matched comparison is attributable to the new fusion; the larger comparison against older packages also benefits from the improved source and generic elimination.

All variables in this report are independently quantified over $\mathbb{Z}$. In particular, no nonnegativity condition on a free coordinate is silently assumed. The coefficient data are fixed integers. The final polynomials use only addition,

subtraction and multiplication, with fixed finite index sets. Exponentiation by a variable occurs in the *proof of correctness*, never in the definition of a delivered equation.

The main contribution is a reusable compression theorem, followed by an explicit instantiation. It does not introduce a new formalization of set theory. The selected source encodes the finite numerical calculus C222 directly, and its arithmetic calibration connects C222 refutations to ZF and ZFC refutations. The present polynomial is therefore not a simulation of K215 and does not instantiate Jones's universal polynomial. Those distinctions concern the route used, not an assertion that their other tradeoffs have become inferior.

There is no agreed scalar measure of "simplest." Variable count, total degree, coefficient height, number of monomials, straight-line-program length, and the size of fixed coefficient data compete. The repository already includes a quartic in 37 unrestricted integers, nine-positive- variable equations of much higher degree, and ten-integer equations using a different reduction [5]. Our contribution concerns the direct C222 route with small literal integer data. It does not replace those constructions or establish a lower bound on any unrestricted notion of simplicity.

1.1 Comparison boundary and attribution

The repository snapshot is

`ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b`.

The corpus was inspected through package 172. The main README's summary lags some delivered packages; our arithmetic source is the actual pre-row-reduction 216-coordinate system in package 172, not that summary. The exact source bytes have SHA-256

`e5e14bc0bbabb625f78ce57017fbdb336c4a65c5877f517142c399769992ff39`.

Package 158 introduced pooling at one nonzero base. Package 168 combined the two consecutive-base pools by a reciprocal sum, relying on their common modulus sign, and discussed orientation weights as a possible generalization [3, 2]. Package 172 supplied the present arithmetic source, including its signed bootstrap, exact projections, and ordinary C222 calibration [1]. Package 152 supplied the general monic-elimination method used for the later degree/variable tradeoffs [4]. We credit all of these ingredients.

The new mathematical step is the unweighted pooling theorem below, including mixed signs and zero bases, and its explicit composition with the authenticated source. The search for elimination orders is a finite optimization of that construction. No literature-wide priority claim is made. Independent reviews here mean separately assigned AI-assisted mathematical and program checks, not outside human refereeing.

2 The arithmetic interface

For an integer A, let the canonical signed Pell coordinates be defined by

$$(A+\sqrt{A^2-1})^k = X_k(A) + Y_k(A)\sqrt{A^2-1}, \qquad k \in \mathbb{Z}.$$

This notation can equivalently be defined by integer recurrence, avoiding any analytic choice of a square root. Put

$$M_A = \begin{pmatrix} A & A^2-1 \\ 1 & A \end{pmatrix}, \qquad \begin{pmatrix} X_k(A) \\ Y_k(A) \end{pmatrix} = M_A^k \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The determinant is one, so negative powers also have integer entries. In particular $X_{-k} = X_k$ and $Y_{-k} = -Y_k$. Both signs of A are permitted throughout the proof.

Suppose that s bases $n_1, \dots, n_s$ support a total of r requests. For $i \in I_j$, the proposed request is $m_i = n_j^{k_i}$ and the

retained Pell pair is $(X_i, Y_i) = (X_{k_i}(A), Y_{k_i}(A))$. The sets I_j are fixed finite index sets. Define

$$\begin{aligned} T_j &= 2An_j - n_j^2 - 1, \qquad (1)\\ \Delta_i &= m_i - X_i + AY_i,\\ H_j &= \sum_{i\in I_j} Y_i(m_i - X_i),\\ S_j &= \sum_{i\in I_j} (\Delta_i^2 - Y_i^2),\\ E_j &= S_j - 2n_j H_j. \qquad (2) \end{aligned}$$

In the literal source, $Z_i = AY_i$ is an already retained coordinate or affine expression. Thus $\Delta_i = m_i - X_i + Z_i$ is affine, H_j, S_j are quadratic, and E_j is at most cubic. The letters in these displays are polynomial abbreviations unless explicitly declared as coordinates.

Lemma 2.1 (Pell congruence for signed indices)**.** *If $n \neq 0$ and $T = 2An - n^2 - 1$, then $\gcd(n, T) = 1$. Whenever $T \neq 0$,*

$$X_k(A) + (n - A)Y_k(A) \equiv n^k \pmod{T} \qquad (k \in \mathbb{Z}),$$

where negative exponents on the right mean powers of the unit n in $\mathbb{Z}/T\mathbb{Z}$.

Proof. The identity $T \equiv -1 \pmod{n}$ proves coprimality. For the row vector $\ell = (1, n - A)$, direct multiplication gives

$$\ell M_A = (n, An - 1) \equiv n(1, n - A) = n\ell \pmod{T},$$

because $An - 1 - n(n - A) = T$. Both M_A and n are invertible modulo T, so this relation iterates through every signed integer k. Apply it to $(1, 0)^{\mathsf{T}}$. No sign or parity assumption on A or k has been used. □

Write $\delta_i = m_i - X_i + (A - n_j)Y_i$ for $i \in I_j$. Expansion gives the central identity

$$\sum_{i\in I_j} \delta_i^2 = E_j - T_j \sum_{i\in I_j} Y_i^2. \qquad (3)$$

The symbolic checker verifies this identity using only the retained $Z_i - AY_i$ equations, with all actual coordinate offsets included.

3 Unweighted pooling across arbitrary signed bases

Theorem 3.1 (Two integrality steps)**.** *Let $W \geq 258$ and $|A| > W^W$, and put $b = \lfloor \sqrt{W - 1} \rfloor$. Suppose*

$$n_j^2 < W, \qquad m_i^2 < W, \qquad k_i^2 < W,$$

and suppose the Pell pairs are canonical at their stated signed indices. Assume also

$$4r < W, \qquad (s + 1)b + \left\lceil \frac{s + 1}{2} \right\rceil + 1 \leq W. \qquad (4)$$

The following conditions are equivalent:

1. *$T_j \mid E_j$ for every j;*
2. *every request at a nonzero base is the exact rational equality $m_i = n_j^{k_i}$;*
3. *there is $z \in \mathbb{Z}$ such that*

$$\boxed{R = \sum_{j=1}^{s} E_j \prod_{h\neq j} T_h - z \prod_{h=1}^{s} T_h = 0.} \qquad (5)$$

For $|n_j| \geq 2$, the same conditions force $k_i \geq 0$. Zero bases, unit bases, repeated bases, and noncoprime moduli are allowed. In particular, the two budget inequalities hold uniformly for $s \leq 14$, $r \leq 64$.

Proof. First note that every T_j is nonzero. If $n_j = 0$, then $T_j = -1$. Otherwise, since $1 \le |n_j| \le b$,

$$|T_j| \ge |n_j|(2|A| - W) > |n_j|W^W, \qquad \operatorname{sgn}(n_j T_j) = \operatorname{sgn}(A). \tag{6}$$

Indeed $|n_j| + 1/|n_j| \le b + 1 \le W$, and $|A| > W^W > W$.

Assume (5). Divide by the nonzero product of moduli and use (3). Then $\sum_{j,i\in I_j} \delta_i^2/T_j$ is an integer. Terms at zero bases are already integers, because their denominators equal -1. Remove them from this rational sum. If no bases remain, every required divisibility is automatic. Otherwise let $J = \{j : n_j \ne 0\}$ and set

$$L = \prod_{j\in J} n_j, \qquad C = L^b \ne 0, \qquad d_i = Cm_i - \frac{C}{n_j^b} n_j^{b+k_i} \quad (i \in I_j,\ j \in J).$$

These are integers: $b + k_i \ge 0$ and C/n_j^b is a product of integer powers. Lemma 2.1 implies $C\delta_i \equiv d_i \pmod{T_j}$, including negative indices. Consequently, with

$$D_j^* = \sum_{i\in I_j} d_i^2, \qquad D^* = \sum_{j\in J} D_j^*,$$

the rational number $R_0 = \sum_{j\in J} D_j^*/T_j$ differs by an integer from $C^2 \sum \delta_i^2/T_j$. Hence $R_0 \in \mathbb{Z}$.

Every input has absolute value at most b. If $s' = |J| \le s$, then

$$|d_i| \le b^{s'b+1} + b^{(s'+1)b} \le 2b^{(s+1)b}, \qquad D^* \le 4rb^{2(s+1)b}.$$

The stronger estimate needed momentarily is

$$\begin{aligned} 2|L|D^* &\le 8rb^{2(s+1)b+s} \le 4rb^{2(s+1)b+s+1} \\ &< W^{(s+1)b+\lceil (s+1)/2\rceil+1} \le W^W. \end{aligned} \tag{7}$$

Here $2 \le b$, $b^2 < W$, $4r < W$, and (4) are sufficient. In particular $D^* < W^W$. Thus (6) gives $|R_0| \le D^*/(2|A|-W) < 1$. Since R_0 is an integer, $R_0 = 0$.

This is not yet enough: denominators can have opposite signs. The second integrality step uses the exact identity

$$\frac{2A}{T_j} = \frac{1}{n_j} + \frac{n_j^2 + 1}{n_j T_j}. \tag{8}$$

Multiplying $R_0 = 0$ by $2A$ gives $U + V = 0$, where

$$U = \sum_{j\in J} \frac{D_j^*}{n_j}, \qquad V = \sum_{j\in J} \frac{D_j^*(n_j^2+1)}{n_j T_j}.$$

Now $LU \in \mathbb{Z}$, so $LV = -LU \in \mathbb{Z}$. On the other hand,

$$\frac{n_j^2+1}{|n_j T_j|} \le \frac{1 + n_j^{-2}}{2|A| - W} \le \frac{2}{2|A| - W},$$

and (7) yields $|LV| \le 2|L|D^*/(2|A| - W) < 1$. Therefore $LV = 0$ and $V = 0$. Every nonzero summand of V has the same sign, namely $\operatorname{sgn}(A)$. All D_j^* and all d_i must vanish. Dividing $d_i = 0$ in $\mathbb{Q}$ by C proves $m_i = n_j^{k_i}$. For a nonunit base and a negative index, this would imply $m_i n_j^{-k_i} = 1$, which is impossible in $\mathbb{Z}$.

Conversely, exact powers at nonzero bases and Lemma 2.1 give $T_j \mid \delta_i$. Identity (3) then gives $T_j \mid E_j$. At a zero base this divisibility is automatic. Finally, if every $T_j \mid E_j$, choose $z = \sum_j E_j/T_j \in \mathbb{Z}$ to obtain (5).

For the uniform numerical corollary, $b \ge 16$ and

$$(s+1)b + \lceil (s+1)/2\rceil + 1 \le 15b + 9 \le b^2 + 1 \le W$$

when $s \le 14$. Also $4r \le 256 < W$ when $r \le 64$. □

3.1 Why the second step matters

A single small-integer argument only proves $R_0 = 0$. It does not prevent cancellation between moduli of opposite signs. The decomposition (8) separates a rational leading term with controlled denominator from a smaller correction whose sign is common. This is the additional mechanism that permits removal of the orientation factors.

The height assumption is essential. Set $A = 10$ and let every index be zero, so every canonical pair is $(1, 0)$. At base $+1$ take proposed output 4, giving $E_+ = 9, T_+ = 18$. At base -1 take outputs $4, 2, 2$, giving $E_- = 11, T_- = -22$. Then

$$E_+/T_+ + E_-/T_- = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

although neither divisibility holds. This example violates the large- center hypothesis. It is included in the independent audit as a concrete reason not to delete that hypothesis.

3.2 Integral witness recovery

The retained source uses two kinds of power block. An individual request has private coordinates T, Q and rows

$$T - (2An - n^2 - 1) = 0, \qquad \delta + TQ = 0.$$

After applying the theorem, recover

$$T = 2An - n^2 - 1, \qquad Q = -\delta/T \in \mathbb{Z}.$$

At a nonzero base, exact powers guarantee the required divisibility. At a zero base, $T = -1$ guarantees it without interpreting 0^k.

A same-base block has private K, L and rows

$$K - 2H - (2A - n)L = 0, \qquad S - nK + L = 0. \tag{9}$$

Restore

$$L = E/T, \qquad K = 2H + (2A - n)L.$$

Both are integers and satisfy (9). For $n = 0$, simply $L = -E$. Thus the new equation restores the full old polynomial system before the old semantic bootstrap is invoked. No circular argument about nonzero bases is introduced.

These maps preserve the set of retained tuples that admit witnesses. They are not polynomial coordinate bijections on all of affine space: division is used only in the proof of integral recovery on the zero set. The emitted polynomial itself contains no division.

4 The explicit calibrated source and its transformation

4.1 The five blocks

The literal source `S216` has 216 unrestricted integer coordinates $x_0, \ldots, x_{215}$ and 153 quadratic residuals $r_0, \ldots, r_{152}$. Appendix B prints every residual. The arithmetic roles used here are

$$A = 1 + x_{133}, \quad t = x_0, \quad F = x_3, \quad B = x_4, \quad BH = x_{185}, \quad q = 1 + 4x_{209}.$$

The five bases, in their fixed order, are

$$(n_0, n_1, n_2, n_3, n_4) = (t, B, BH, q + 1, q). \tag{10}$$

The first three blocks each have one request. The fourth has eleven and the fifth has three, for a total of seventeen. The complete request table appears in Appendix A. Repeated use of a Pell pair at the same index does not create a new coordinate.

Delete exactly the ten rows

$$\mathcal{D} = \{97, 98, 99, 100, 101, 102, 149, 150, 151, 152\} \tag{11}$$

and the ten coordinates

$$\mathcal{V} = \{140, 141, 142, 143, 144, 145, 212, 213, 214, 215\}. \tag{12}$$

The first six coordinates are the three individual moduli and quotients; the last four are the two K, L pairs. They occur nowhere in the retained 143 rows. Their exact private occurrences and their signs are checked.

Define the five T_j, H_j, S_j, E_j from the complete request table and (1)–(2), and let R be (5) with $s = 5$. The first explicit new polynomial is

$$\boxed{P_{207} = \sum_{i \notin \mathcal{D}} r_i^2 + R^2.} \tag{13}$$

Its free coordinates are the 206 old coordinates outside $\mathcal{V}$ and the new integer z. For software, the retained old indices are renumbered in increasing order, followed by z; the exact map is stored.

Because every base and A is affine, each T_j is quadratic and each E_j is at most cubic. Thus R has degree at most 11. Its coefficient of the private variable z is $-\prod_j T_j$, a polynomial of degree exactly ten. Each T_j has a nonzero quadratic homogeneous part, and $\mathbb{Z}[\mathbf{x}]$ is an integral domain. Therefore $\deg R = 11$ and $\deg P_{207} = 22$ exactly. Taking sums of squares introduces no cancellation of the highest nonzero homogeneous square.

4.2 Why the retained hypotheses still hold

The source proof establishes canonical signed Pell pairs and the large center before interpreting the power congruences or recovering front signs. The actual frame is

$$W = 2 + \sum_{a \in \mathcal{A}} a^2, \qquad |\mathcal{A}| = 34, \tag{14}$$

where the complete affine-atom list is supplied in `inputs/interface216.json`. The independent checker expands (14) and matches source row 129 exactly. Every actual base, output and index of all seventeen requests is one of these atoms. The t atom and retained horizon equation give $t \geq 16$, hence $W \geq 258$. The retained scale and semantic-anchor argument gives $|A| > W^W$ [1].

All fourteen $Z_i = AY_i$ equations, fourteen Pell norm equations, the complete signed-orientation equation, the scale, and the frame remain. The checker also reconstructs source row 142 as the orientation identity minus the fourteen retained norm identities. Three delicate conventions are checked literally: A is stored with offset $+1$; pair 3's ordinate has offset -1; pair 12's Z has offset $+1$. A variable name alone is never taken as evidence of its arithmetic value.

Here $s = 5, r = 17$, so $4r = 68 < 258$ and $6b + 4 \leq b^2 + 1 \leq W$. Theorem 3.1 applies to every zero of the new retained system. It restores all ten deleted witnesses and all ten source rows. Conversely, every source zero supplies an integer z. Thus P_{207} and `S216` have equivalent integer-zero existence.

4.3 The logical endpoint

The ordinary source calibration proves

$$\begin{aligned} \exists \mathbf{x} \in \mathbb{Z}^{216}\ (r_i(\mathbf{x}) = 0 \text{ for all } i) \quad &\iff \quad \text{there is a finite C222 refutation} \\ &\iff \quad \text{there is a finite ZF refutation} \\ &\iff \quad \text{there is a finite ZFC refutation.} \end{aligned}$$

The named logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

Their provenance and source files are retained with the parent arithmetic proofs. The appearance of the name `KEndpoint` in an inherited interface is not a Turing-machine simulation in the present polynomial.

All new steps are finite integer algebra, elementary bounds, recurrence arguments, and explicit finite-witness transformations. They are formalizable in ZFC and do not assume Con(ZFC). Composing with the identified ordinary source calibration gives

$$\mathrm{ZFC} \vdash \left(\forall \mathbf{u} \in \mathbb{Z}^{207},\ P_{207}(\mathbf{u}) \neq 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{15}$$

Here Con(ZFC) refers to absence of a finite refutation in the effective presentation fixed by the inherited calibration. No semantic soundness assumption on ZFC is needed to state or prove the equivalence. The report does not prove either side of it.

5 Removing defined coordinates

Lemma 5.1 (Acyclic monic elimination)**.** *Suppose selected residuals have the form $r_j = \varepsilon x_v + f$, where $\varepsilon \in \{1, -1\}$ and f does not contain x_v. If their dependency graph is acyclic, replace every selected x_v by the recursively defined polynomial $-\varepsilon f$, omit the selected residuals, and retain every other residual after substitution. This gives a polynomial bijection between the zero sets in the remaining and original coordinates.*

Proof. Resolve the definitions in a topological order. Each gives one unique integer coordinate for any values of its already resolved arguments. The omitted row then vanishes identically. The retained rows become exactly the substituted rows. Forward restriction and reverse polynomial restoration are mutual inverses on the zero sets. □

This lemma is a standard exact elimination step and was already applied extensively in package 152. The contribution here is its composition with the new power pooling and the explicit selected pivot tables. Every pivot is checked against a retained *original* source row: its coefficient is exactly $+1$ or -1, it has no other occurrence in that row, and no private power coordinate is referenced. The compiler checks acyclicity and recursively substitutes the definitions. No rational Gaussian-elimination step is used to remove an integer coordinate.

The principal smaller polynomials are fully defined by (13), or by its specified block partition, followed by the finite pivot lists in Appendix C. A pivot (i, v) always means: solve the printed original row r_i for x_v with its unit coefficient. Resolve all such definitions recursively, remove their rows, and sum the squares of the remaining substituted residuals and the specified pooled residuals. This is a complete polynomial recipe with fixed integer coefficients; it does not call an unspecified Diophantine compiler.

5.1 Partitioning blocks to control degree

The five blocks may be partitioned into fixed subsets. Apply Theorem 3.1 separately to each subset, introducing one z per part. The unchanged square-frame budget is sufficient for every part. Before monic eliminations, the selected partitions have these exact counts:

Block sizes	Integer variables	Residuals	Degree	Height bound
5	207	144	22	$< 2^{46}$
$3+2$	208	145	14	$< 2^{29}$
$2+2+1$	209	146	10	$< 2^{25}$
$1+1+1+1+1$	211	148	6	$< 2^{24}$

Even a singleton part can replace its two old private coordinates by one witness, at a degree cost. These rows are intermediate constructions, not a claim that their raw variable counts improve the repository's already eliminated high-degree equations.

5.2 Exact degree, liveness, and coefficient size

The emitted straight-line programs are complete, topologically ordered arrays with integer leaves and binary $+, -, \times$ gates. Their inputs are exactly the declared free integer coordinates. Sharing a node is only an evaluator optimization and adds no mathematical variable.

Degrees are certified in two steps. Propagating syntactic degrees gives an upper bound. Propagating homogeneous leading values at a supplied integer point gives a nonzero value in that upper degree, proving equality. The independent checkers recompute these witnesses. Nonzero partial derivatives at explicit finite-field points certify that every declared coordinate really occurs in the final polynomial; syntactic appearance alone would not prove this after cancellation.

We have not expanded the largest output polynomials, so exact monomial counts and exact expanded coefficient heights are not claimed. A useful rigorous bound is nevertheless available. Assign norm bound 1 to each variable, $|c|$ to an integer constant, sum bounds at addition or subtraction, and multiply bounds at multiplication. These rules bound the coefficient ℓ^1 norm, hence every expanded coefficient. If the resulting positive integer has bit length B, then $\mathrm{Ht}(P) < 2^B$. The reported bounds use the complete emitted circuits, including the sum of residual squares. The largest literal integer in those circuits is 60; this must not be confused with their expanded coefficient height.

6 Verification, reproducibility, and remaining limits

The release contains two independent mathematical reviews of the mixed-sign pooling argument. The first supplies the uniform theorem and the signed Pell congruence proof; the second separately checks the five-base/seventeen-request bounds, zero and unit bases, and the small-center counterexample. The independent finite checks cover:

- the immutable source hash, all seventeen request ports, the exact 34-atom frame, all fourteen Pell pairs, offsets, and every private occurrence of the ten removed coordinates;
- the new T, H, S, E expressions, the error-square identity, pool partitions, the complete retained residual list, and emitted circuits;
- every monic pivot, its integer inverse definition, acyclicity, the full substituted circuit, exact degree and variable liveness;
- arithmetic samples with both signs of A, mixed and zero bases, units with negative indices, false outputs, and impossible negative nonunit powers, together with deliberate corruptions;
- deterministic reconstruction in a fresh directory and both ordinary and optimized Python execution, so disabled `assert` statements cannot silently remove the validation.

The restored package-172 source tree and its finite arithmetic ancestry were also replayed. Its recorded receipts distinguish that replay from a new Lean/Rocq build, which was not performed. The full ordinary proof of the inherited calibration is a dependency; finite checks do not turn it into a proof-assistant certificate. Similarly, numerical tests of the new arithmetic theorem support implementation review but do not replace the quantified proof in Section 3.

Run the release verifier from any working directory using an absolute path to `verify.py`. The checker requires Python 3.10 or later and its standard library. `evaluate.py` evaluates a selected literal circuit on one supplied tuple, exactly or modulo a chosen integer. It performs no root search. A modular zero has no asserted relationship to a ZFC refutation.

The verification record in `verification/` identifies exactly what was run. `SOURCES.json` and `NOTICE.md` distinguish new results from inherited source files. The archive's manifest covers the final release bytes. The LaTeX sources and all generated inputs are present; an ordinary `pdflatex` installation rebuilds this report. Rebuilding may change PDF metadata without changing its content.

7 Further research questions

Can the two-scale argument reduce more shared certificates? The proof exploits the particular expansion $2A/T = 1/n + (n^2+1)/(nT)$. More generally, one can seek denominator families with a rational leading term of bounded denominator and a correction of one sign. A useful extension would state precise height conditions and produce a polynomial with a smaller aggregate degree. Merely taking an arbitrary reciprocal sum is invalid, as the explicit small-center counterexample demonstrates.

How far can partition-aware elimination go? The selected orders are finite successful searches, not optima. The choice of block partition interacts with which bases, moduli, and Pell products may be inlined while keeping a target degree. A joint search could use exact degree certificates during optimization rather than only safe

weighted-degree upper bounds. Any new point should be compared against the same source with the fusion disabled, so the contribution of the arithmetic theorem remains measurable.

Can the quotient witness be carried by another coordinate? The new aggregate adds one unrestricted integer per part. If a retained coordinate has a demonstrably free cofinal choice, a polynomial carrier might encode this quotient without adding a new coordinate. Such a proposal needs both an integral soundness recovery and a constructive completeness argument. No such carrier is claimed here.

Can a comparable gain be obtained at degree four? The present compression deliberately spends degree. Returning to a quartic requires lifting products to new coordinates, which may erase the variable savings. A useful objective is a structural identity that retains the cancellation argument but realizes its conclusion with fewer quadratic constraints than the original five blocks. The current result supplies no lower bound ruling this out.

Can the calibration and the general theorem be formalized together? A focused formal development could first prove the rational two- integrality lemma, then its signed-Pell specialization, and finally the literal source transformation. That is smaller in scope than rebuilding an entire universal Diophantine theorem. The source arithmetic compiler and object-language ZFC formalization boundary would still need to be tracked explicitly. We have not represented ordinary formalizability as a completed kernel check.

What is the best presentation under a declared simplicity budget? An exact comparison should fix the domain and charge for coefficient data, degree, and either expanded support or a specified arithmetic circuit. The direct route here avoids a large native coefficient tuple but uses more variables than the repository's other routes. Proving a global minimum is far beyond the current evidence. A reproducible Pareto catalog with matched controls would be a concrete next result.

A Complete request interface

The following table is literal data, not an informal description. The source coordinate numbers are used throughout. The quantities $X = D + Z$, n, m and k supply the definitions in Section 2. The final request uses the affine expression

$$J = 5x_8 - 3x_9 + 4x_{19} + 2x_{22} + 2x_{26} + 2x_{28} + 2x_{30} + 2x_{32} + 4x_{34} + 2x_{35} + 2x_{38} + 2x_{41} + 4x_{60} + 2x_{61}.$$

Request	Block	n	m	k	$D = X - AY$	(Y, Z)
0	0	x_0	x_3	x_0	x_{137}	(x_{138}, x_{139})
1	1	x_4	x_{17}	x_1	x_{173}	$(x_{174}, 1 + x_{175})$
2	2	x_{185}	$1 + 4x_{209}$	x_1	x_{173}	$(x_{174}, 1 + x_{175})$
3	3	$2 + 4x_{209}$	x_{94}	$x_0 + x_3$	x_{146}	(x_{147}, x_{148})
4	4	$1 + 4x_{209}$	x_{95}	x_0	x_{137}	(x_{138}, x_{139})
5	3	$2 + 4x_{209}$	x_{99}	$-1 + x_8$	x_{149}	$(-1 + x_{150}, x_{151})$
7	3	$2 + 4x_{209}$	x_{102}	x_9	x_{152}	(x_{153}, x_{154})
8	3	$2 + 4x_{209}$	x_{105}	x_{10}	x_{155}	(x_{156}, x_{157})
9	3	$2 + 4x_{209}$	x_{108}	x_{11}	x_{158}	(x_{159}, x_{160})
10	3	$2 + 4x_{209}$	x_{111}	x_{12}	x_{161}	(x_{162}, x_{163})
11	4	$1 + 4x_{209}$	x_{112}	x_2	x_{164}	(x_{165}, x_{166})
12	3	$2 + 4x_{209}$	x_{115}	x_{13}	x_{167}	(x_{168}, x_{169})
13	3	$2 + 4x_{209}$	x_{118}	x_{14}	x_{170}	(x_{171}, x_{172})
14	4	$1 + 4x_{209}$	x_{119}	x_1	x_{173}	$(x_{174}, 1 + x_{175})$
15	3	$2 + 4x_{209}$	x_{123}	x_{15}	x_{176}	(x_{177}, x_{178})
16	3	$2 + 4x_{209}$	x_{126}	x_{16}	x_{179}	(x_{180}, x_{181})
17	3	$2 + 4x_{209}$	x_{129}	J	x_{182}	(x_{183}, x_{184})

B Every quadratic source residual

Set each of the following 153 polynomials equal to zero to recover the literal source S216. Row numbers are zero-based. The rows in (11) are removed only by the proved pooling transformation. No other row is silently dropped. These formulas, the request table and the pivot tables specify every final polynomial without requiring external numerical data.

$$r_0 = x_0 - x_0x_2 - x_2 + x_7$$

$$r_1 = -1 + x_1 - x_2x_5 - x_3$$

$$r_2 = -1 + x_1 - x_6x_7$$

$$r_3 = 1 + x_3 - x_4x_{18} + x_{16}$$

$$r_4 = -1 + x_8 - x_8^2 - 2x_8x_9 - x_9 - x_9^2 + x_{19}$$

$$r_5 = -5 - 20x_8 - 16x_8^2 - 16x_8x_9 - 16x_8x_{19} - 10x_9 - 4x_9^2 - 8x_9x_{19} - 10x_{19} - 4x_{19}^2 + x_{20}$$

$$r_6 = -2 - x_8 - x_8^2 - 4x_8x_{20} - 4x_{20} - 4x_{20}^2 + x_{21}$$

$$r_7 = 1 + 2x_{21} - 4x_{21}^2 + x_{22}$$

$$r_8 = -2 - x_9^2 - 2x_9x_{10} - x_{10} - x_{10}^2 + x_{23}$$

$$r_9 = -1 + x_8 - x_8^2 - 2x_8x_{10} - x_9 - x_{10}^2 + x_{24}$$

$$\begin{aligned} r_{10} = 8 &- 11x_8 - 49x_8^2 + 10x_8x_9 - 44x_8x_{10} - 32x_8x_{19} + 4x_8x_{20} - 4x_8x_{23} - 32x_8x_{24} + 25x_9 + 3x_9^2 \\ &+ 8x_9x_{10} - 8x_9x_{23} - 15x_{10} - 12x_{10}^2 - 16x_{10}x_{19} + 8x_{10}x_{23} - 16x_{10}x_{24} - 9x_{19} - 4x_{19}^2 - 8x_{19}x_{24} \\ &+ 4x_{20} + 4x_{20}^2 - x_{21} + 12x_{23} - 4x_{23}^2 - 8x_{24} - 4x_{24}^2 + x_{25} \end{aligned}$$

$$\begin{aligned} r_{11} = -18 &+ 14x_8 - 2x_8^2 - 8x_8x_9 + 8x_8x_{10} - 8x_8x_{23} + 24x_9 - 8x_9^2 + 16x_9x_{10} - 16x_9x_{23} - 30x_{10} - 44x_{10}^2 \\ &+ 16x_{10}x_{23} + 24x_{10}x_{25} + 24x_{23} - 8x_{23}^2 - 4x_{25}^2 + x_{26} \end{aligned}$$

$$\begin{aligned} r_{12} = -6 &- 8x_8 - 4x_8^2 - 8x_8x_9 - 8x_8x_{10} + 8x_8x_{23} + 8x_8x_{24} - 6x_9 - 4x_9^2 - 8x_9x_{10} + 8x_9x_{23} + 8x_9x_{24} \\ &- 5x_{10} - 4x_{10}^2 + 8x_{10}x_{23} + 8x_{10}x_{24} - x_{19} + 8x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 6x_{24} - 4x_{24}^2 + x_{27} \end{aligned}$$

$$r_{13} = 1 - 4x_8 + 2x_9 - 4x_9^2 - 8x_9x_{10} - 8x_9x_{27} + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{27} - 2x_{19} + 4x_{27} - 4x_{27}^2 + x_{28}$$

$$\begin{aligned} r_{14} = 35 &- 16x_8 + 2x_8^2 + 18x_8x_9 - 12x_8x_{10} - 20x_8x_{23} - 32x_8x_{24} - 8x_9 - 14x_9^2 - 24x_9x_{10} - 40x_9x_{23} \\ &- 12x_9x_{24} - 26x_{10} - 24x_{10}^2 - 40x_{10}x_{23} - 32x_{10}x_{24} - x_{19} - 40x_{23} + 4x_{23}^2 + 16x_{23}x_{24} - 28x_{24} + 4x_{24}^2 \\ &- 2x_{27} + x_{29} \end{aligned}$$

$$\begin{aligned} r_{15} = 8 &+ 34x_8 + 18x_8^2 - 8x_8x_9 + 24x_8x_{10} + 8x_8x_{23} + 32x_8x_{24} - 50x_9 + 10x_9^2 - 32x_9x_{10} + 16x_9x_{23} \\ &- 8x_9x_{24} + 60x_{10} - 4x_{10}^2 - 16x_{10}x_{23} + 32x_{10}x_{24} - 24x_{10}x_{25} - 32x_{10}x_{29} - 24x_{23} + 8x_{23}^2 + 32x_{24} \\ &+ 8x_{24}^2 + 4x_{25}^2 - x_{26} - 2x_{29} - 4x_{29}^2 + x_{30} \end{aligned}$$

$$\begin{aligned} r_{16} = 6 &- 2x_8 + x_8^2 + 4x_8x_{20} - 47x_9 - 28x_9^2 + 16x_9x_{10} + 24x_9x_{19} + 24x_9x_{23} + 7x_{10} + 8x_{10}^2 + 16x_{19} \\ &- 4x_{19}^2 - 8x_{19}x_{23} + 4x_{20} + 4x_{20}^2 - x_{21} + 6x_{23} - 4x_{23}^2 - x_{24} + x_{31} \end{aligned}$$

$$r_{17} = -3 - 6x_8 + 2x_9 - 2x_{10} - 2x_{24} - 4x_{31} - 4x_{31}^2 + x_{32}$$

$$\begin{aligned} r_{18} = 16 &+ 30x_8 + 14x_8^2 - 2x_8x_9 - 24x_8x_{10} - 16x_8x_{19} + 60x_8x_{20} - 16x_8x_{27} + 19x_9 - 6x_9x_{10} - 24x_9x_{19} \\ &+ 32x_9x_{20} - 12x_{10} + 6x_{10}^2 - 12x_{10}x_{19} + 12x_{10}x_{27} + 3x_{19} - 8x_{19}^2 + 32x_{19}x_{20} - 8x_{19}x_{27} + 15x_{20} \\ &- 4x_{20}^2 - x_{23} - 2x_{24} + 4x_{27}^2 + x_{33} + 8x_{36} - 8x_{36}^2 + 2x_{37} - x_{42} \end{aligned}$$

$$r_{19} = 1 + 2x_{33} - 4x_{33}^2 + x_{34}$$

$$r_{20} = -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35}$$

$$\begin{aligned} r_{21} = -1 &- 15x_8 - 17x_8^2 + 30x_8x_9 - 32x_8x_{19} + 7x_9 - 13x_9^2 + 8x_9x_{10} + 32x_9x_{19} - x_{10} - 14x_{19} - 16x_{19}^2 \\ &- 4x_{23} + x_{24} + x_{36} + 2x_{71} \end{aligned}$$

$$r_{22} = -4 - 3x_8 - 7x_9 - 4x_{10} - 2x_{23} - x_{24} + 4x_{36} - 4x_{36}^2 + x_{37}$$

$$r_{23} = -6x_8 + 2x_9 + 2x_{10} - 4x_{10}^2 - 8x_{10}x_{37} - 2x_{24} + 4x_{37} - 4x_{37}^2 + x_{38}$$

$$r_{24} = -x_{10} - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}$$

$$\begin{aligned} r_{25} = -31 &+ 2x_8 - x_8^2 - 4x_8x_{39} - 22x_9 + 16x_9^2 - 32x_9x_{10} + 32x_9x_{23} - 42x_{10} - 20x_{10}^2 - 8x_{10}x_{23} \\ &+ 24x_{10}x_{24} + 12x_{23}^2 - 8x_{23}x_{24} + 22x_{24} - 4x_{24}^2 + 2x_{39} - 4x_{39}^2 + x_{40} - 4x_{43} \end{aligned}$$

$$r_{26} = 4x_{8} - 2x_{8}^{2} - 8x_{8}x_{39} - 16x_{10}^{2} - 16x_{10}x_{40} + 4x_{39} - 8x_{39}^{2} - 4x_{40}^{2} + x_{41}$$

$$\begin{aligned} r_{27} = & -16x_{8}^{2} + 24x_{8}x_{10} - 16x_{8}x_{19} + 16x_{8}x_{27} + x_{10} - 9x_{10}^{2} + 12x_{10}x_{19} - 12x_{10}x_{27} - 4x_{19}^{2} + 8x_{19}x_{27} \\ & - 4x_{27}^{2} + x_{42} \end{aligned}$$

$$r_{28} = -1 + 6x_{9} - 4x_{9}^{2} + 8x_{9}x_{10} - 8x_{9}x_{23} - 6x_{10} - 4x_{10}^{2} + 8x_{10}x_{23} + 6x_{23} - 4x_{23}^{2} + x_{43}$$

$$r_{29} = -1 - x_{10} - x_{10}^{2} - 4x_{10}x_{43} - 4x_{43}^{2} + x_{44}$$

$$r_{30} = 1 + x_{9} - x_{9}^{2} - 4x_{9}x_{44} - 4x_{44}^{2} + x_{45}$$

$$r_{31} = -1 - 6x_{45} - 4x_{45}^{2} + x_{46}$$

$$r_{32} = -1 - x_{10} - x_{10}^{2} - 4x_{10}x_{46} - 4x_{46}^{2} + x_{47}$$

$$r_{33} = 2 + x_{9} - x_{9}^{2} - 4x_{9}x_{47} - 4x_{47}^{2} + x_{48}$$

$$r_{34} = -2x_{42} - 4x_{42}^{2} - 8x_{42}x_{48} - 4x_{48}^{2} + x_{49}$$

$$r_{35} = -9 + 7x_{8} - x_{8}^{2} - 4x_{8}x_{49} + 12x_{49} - 4x_{49}^{2} + x_{50}$$

$$r_{36} = -1 - 6x_{50} - 4x_{50}^{2} + x_{51}$$

$$r_{37} = -x_{10} - x_{10}^{2} - 4x_{10}x_{51} - 4x_{51}^{2} + x_{52}$$

$$r_{38} = -x_{8} - x_{8}^{2} - 4x_{8}x_{52} - 4x_{52} - 4x_{52}^{2} + x_{53}$$

$$r_{39} = -10 - x_{9} + x_{10} - x_{23} - 14x_{53} - 4x_{53}^{2} + x_{54}$$

$$r_{40} = 1 - 2x_{9} + 2x_{10} - 2x_{23} + 4x_{54} - 4x_{54}^{2} + x_{55}$$

$$r_{41} = -10 + x_{9} - x_{9}^{2} - 4x_{9}x_{55} - 14x_{48} - 4x_{48}^{2} - 4x_{55}^{2} + x_{56}$$

$$r_{42} = 21 + 28x_{48} + 8x_{48}^2 - 2x_{56} - 4x_{56}^2 + x_{57}$$

$$r_{43} = 15x_9 - 36x_9^2 + 12x_9x_{10} + 24x_9x_{19} - x_{10} - x_{10}^2 - 4x_{10}x_{19} - 5x_{19} - 4x_{19}^2 + x_{58}$$

$$r_{44} = 1 + 6x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} - 4x_{57}^2 + 4x_{58} - 4x_{58}^2 + x_{59}$$

$$r_{45} = -2 - 2x_{10} + 2x_{10}^2 + 8x_{10}x_{57} + 8x_{57}^2 - 2x_{59} - 4x_{59}^2 + x_{60}$$

$$r_{46} = -2 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$r_{47} = -x_3x_{13} + x_{62}$$

$$r_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$r_{49} = 12x_{10}x_{62} + x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$r_{50} = x_{12} - x_{13} - 2x_{28}x_{62} - 6x_{62} - x_{65}$$

$$r_{51} = -12x_{10}x_{62} + x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$r_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67}$$

$$r_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$r_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$r_{55} = x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70}$$

$$r_{56} = -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{71}$$

$$r_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$

$$r_{58} = -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 + x_{73} - x_{74}^2$$

$$r_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{74}$$

$$r_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$r_{61} = -1 + 2x_3 + 6x_3x_9 - 2x_3x_{19} + x_{75}$$

$$r_{62} = -x_8x_{211} - x_{12} + x_{76} + x_{211}$$

$$r_{63} = -4x_8x_{62} - 2x_9x_{62} + x_9x_{211} + x_{12} - x_{13} - 2x_{19}x_{62} + x_{78}$$

$$r_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$r_{65} = -x_{64}x_{79} + x_{80}$$

$$r_{66} = -x_{65}x_{80} + x_{81}$$

$$r_{67} = -x_{66}x_{81} + x_{82}$$

$$r_{68} = -x_{67}x_{82} + x_{83}$$

$$r_{69} = -x_{68}x_{83} + x_{84}$$

$$r_{70} = -x_{69}x_{84} + x_{85}$$

$$r_{71} = -x_{70}x_{85} + x_{86}$$

$$r_{72} = -x_{77}x_{86} + x_{87}$$

$$r_{73} = -x_{78}x_{87} + x_{88}$$

$$r_{74} = -x_{88}^2 + x_{89}$$

$$r_{75} = -x_2 - x_2x_{16} + x_{16} + x_{90}$$

$$r_{76} = -x_{14}x_{90} + x_{91}$$

$$r_{77} = -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{93} - x_{73}x_{89} - x_{91}^2 - x_{92}^2$$

$$r_{78} = -x_2x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$r_{79} = -x_{96} - 4x_{96}x_{209} + x_{98}$$

$$r_{80} = -x_4x_{119} + x_{94}x_{201} - x_{97} - x_{98}x_{119}$$

$$r_{81} = x_{99}x_{201} - x_{100}x_{122} - x_{101}$$

$$r_{82} = x_{102}x_{201} - x_{103}x_{122} - x_{104}$$

$$r_{83} = x_{105}x_{201} - x_{106}x_{122} - x_{107}$$

$$r_{84} = x_{108}x_{201} - x_{109}x_{122} - x_{110}$$

$$r_{85} = x_{111}x_{202} - x_{113}x_{122} - x_{114}$$

$$r_{86} = x_{115}x_{202} - x_{116}x_{122} - x_{117}$$

$$r_{87} = -x_{17}x_{119} + x_{122}$$

$$r_{88} = x_{118} - x_{120}x_{122} - x_{121}$$

$$r_{89} = -x_{122}x_{124} + x_{123} - x_{125}$$

$$r_{90} = -x_{122}x_{127} + x_{126}x_{201} - x_{128}$$

$$r_{91} = -x_{122}x_{130} + x_{129}x_{201} - x_{131}$$

$$r_{92} = x_{132} - 2x_{132}x_{134} + x_{135}$$

$$r_{93} = -x_{132}x_{135} - x_{135} + x_{136}$$

$$r_{94} = 2x_{133} + x_{133}^2 + x_{135}^2 - x_{136}^2$$

$$r_{95} = -x_{133}x_{138} - x_{138} + x_{139}$$

$$r_{96} = -1 + x_{137}^2 + 2x_{137}x_{139} + x_{138}^2$$

$$r_{97} = 1 - 2x_0 + x_0^2 - 2x_0x_{133} + x_{140}$$

$$r_{98} = -x_0x_{138} + x_3 - x_{137} + x_{140}x_{141}$$

$$r_{99} = 1 - 2x_4 + x_4^2 - 2x_4x_{133} + x_{142}$$

$$r_{100} = -x_4x_{174} + x_{17} + x_{142}x_{143} - x_{173}$$

$$r_{101} = -1 + 2x_{133}x_{185} - x_{144} + 2x_{185} - x_{185}^2$$

$$r_{102} = 1 + x_{144}x_{145} - x_{173} - x_{174}x_{185} + 4x_{209}$$

$$r_{103} = -x_{133}x_{147} - x_{147} + x_{148}$$

$$r_{104} = -1 + x_{146}^2 + 2x_{146}x_{148} + x_{147}^2$$

$$r_{105} = 1 + x_{133} - x_{133}x_{150} - x_{150} + x_{151}$$

$$r_{106} = x_{149}^2 + 2x_{149}x_{151} - 2x_{150} + x_{150}^2$$

$$r_{107} = -x_{133}x_{153} - x_{153} + x_{154}$$

$$r_{108} = -1 + x_{152}^2 + 2x_{152}x_{154} + x_{153}^2$$

$$r_{109} = -x_{133}x_{156} - x_{156} + x_{157}$$

$$r_{110} = -1 + x_{155}^2 + 2x_{155}x_{157} + x_{156}^2$$

$$r_{111} = -x_{133}x_{159} - x_{159} + x_{160}$$

$$r_{112} = -1 + x_{158}^2 + 2x_{158}x_{160} + x_{159}^2$$

$$r_{113} = -x_{133}x_{162} - x_{162} + x_{163}$$

$$r_{114} = -1 + x_{161}^2 + 2x_{161}x_{163} + x_{162}^2$$

$$r_{115} = -x_{133}x_{165} - x_{165} + x_{166}$$

$$r_{116} = -1 + x_{164}^2 + 2x_{164}x_{166} + x_{165}^2$$

$$r_{117} = -x_{133}x_{168} - x_{168} + x_{169}$$

$$r_{118} = -1 + x_{167}^2 + 2x_{167}x_{169} + x_{168}^2$$

$$r_{119} = -x_{133}x_{171} - x_{171} + x_{172}$$

$$r_{120} = -1 + x_{170}^2 + 2x_{170}x_{172} + x_{171}^2$$

$$r_{121} = 1 - x_{133}x_{174} - x_{174} + x_{175}$$

$$r_{122} = -1 + 2x_{173} + x_{173}^2 + 2x_{173}x_{175} + x_{174}^2$$

$$r_{123} = -x_{133}x_{177} - x_{177} + x_{178}$$

$$r_{124} = -1 + x_{176}^2 + 2x_{176}x_{178} + x_{177}^2$$

$$r_{125} = -x_{133}x_{180} - x_{180} + x_{181}$$

$$r_{126} = -1 + x_{179}^2 + 2x_{179}x_{181} + x_{180}^2$$

$$r_{127} = -x_{133}x_{183} - x_{183} + x_{184}$$

$$r_{128} = -1 + x_{182}^2 + 2x_{182}x_{184} + x_{183}^2$$

$$\begin{aligned} r_{129} = & -8 - 2x_0^2 - 2x_0x_3 - x_1^2 - x_2^2 - 2x_3^2 - x_4^2 + 2x_8 - 26x_8^2 + 30x_8x_9 - 40x_8x_{19} - 20x_8x_{22} \\ & - 20x_8x_{26} - 20x_8x_{28} - 20x_8x_{30} - 20x_8x_{32} - 40x_8x_{34} - 20x_8x_{35} - 20x_8x_{38} - 20x_8x_{41} - 40x_8x_{60} \\ & - 20x_8x_{61} - 10x_9^2 + 24x_9x_{19} + 12x_9x_{22} + 12x_9x_{26} + 12x_9x_{28} + 12x_9x_{30} + 12x_9x_{32} + 24x_9x_{34} \\ & + 12x_9x_{35} + 12x_9x_{38} + 12x_9x_{41} + 24x_9x_{60} + 12x_9x_{61} - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ & - x_{15}^2 - x_{16}^2 - x_{17}^2 - 16x_{19}^2 - 16x_{19}x_{22} - 16x_{19}x_{26} - 16x_{19}x_{28} - 16x_{19}x_{30} - 16x_{19}x_{32} \\ & - 32x_{19}x_{34} - 16x_{19}x_{35} - 16x_{19}x_{38} - 16x_{19}x_{41} - 32x_{19}x_{60} - 16x_{19}x_{61} - 4x_{22}^2 - 8x_{22}x_{26} - 8x_{22}x_{28} \\ & - 8x_{22}x_{30} - 8x_{22}x_{32} - 16x_{22}x_{34} - 8x_{22}x_{35} - 8x_{22}x_{38} - 8x_{22}x_{41} - 16x_{22}x_{60} - 8x_{22}x_{61} - 4x_{26}^2 \\ & - 8x_{26}x_{28} - 8x_{26}x_{30} - 8x_{26}x_{32} - 16x_{26}x_{34} - 8x_{26}x_{35} - 8x_{26}x_{38} - 8x_{26}x_{41} - 16x_{26}x_{60} - 8x_{26}x_{61} \\ & - 4x_{28}^2 - 8x_{28}x_{30} - 8x_{28}x_{32} - 16x_{28}x_{34} - 8x_{28}x_{35} - 8x_{28}x_{38} - 8x_{28}x_{41} - 16x_{28}x_{60} - 8x_{28}x_{61} \\ & - 4x_{30}^2 - 8x_{30}x_{32} - 16x_{30}x_{34} - 8x_{30}x_{35} - 8x_{30}x_{38} - 8x_{30}x_{41} - 16x_{30}x_{60} - 8x_{30}x_{61} - 4x_{32}^2 \\ & - 16x_{32}x_{34} - 8x_{32}x_{35} - 8x_{32}x_{38} - 8x_{32}x_{41} - 16x_{32}x_{60} - 8x_{32}x_{61} - 16x_{34}^2 - 16x_{34}x_{35} - 16x_{34}x_{38} \\ & - 16x_{34}x_{41} - 32x_{34}x_{60} - 16x_{34}x_{61} - 4x_{35}^2 - 8x_{35}x_{38} - 8x_{35}x_{41} - 16x_{35}x_{60} - 8x_{35}x_{61} - 4x_{38}^2 \\ & - 8x_{38}x_{41} - 16x_{38}x_{60} - 8x_{38}x_{61} - 4x_{41}^2 - 16x_{41}x_{60} - 8x_{41}x_{61} - 16x_{60}^2 - 16x_{60}x_{61} - 4x_{61}^2 \\ & - x_{94}^2 - x_{95}^2 - x_{99}^2 - x_{102}^2 - x_{105}^2 - x_{108}^2 - x_{111}^2 - x_{112}^2 - x_{115}^2 - x_{118}^2 - x_{119}^2 \\ & - x_{123}^2 - x_{126}^2 - x_{129}^2 + x_{132} - x_{185}^2 - 24x_{209} - 32x_{209}^2 \end{aligned}$$

$$r_{130} = -x_4 - x_4x_8 - x_4x_9 - x_4x_{10} - x_4x_{11} - x_4x_{12} - x_4x_{13} - x_4x_{14} - x_4x_{15} - x_4x_{16} + x_{185}$$

$$r_{131} = -16 + x_0 - x_{186}^2$$

$$r_{132} = -x_{133}x_{188} - x_{188} + x_{197}$$

$$r_{133} = -1 + x_{187}^2 + x_{188}^2 - x_{197}^2$$

$$r_{134} = -x_{190} - x_{190}x_{191} + x_{198}$$

$$r_{135} = -1 + x_{189}^2 + x_{190}^2 - x_{198}^2$$

$$r_{136} = -x_{183}^2 + x_{199}$$

$$r_{137} = x_{188} - 2x_{193}x_{199} - x_{199}$$

$$r_{138} = -4x_{188} - 8x_{188}x_{192} + x_{191}$$

$$r_{139} = -x_{133} - x_{187}x_{194} + x_{191}$$

$$r_{140} = -x_{182} - x_{184} - x_{187}x_{195} + x_{189}$$

$$\begin{aligned} r_{141} = & -5x_8 + 3x_9 - 4x_{19} - 2x_{22} - 2x_{26} - 2x_{28} - 2x_{30} - 2x_{32} - 4x_{34} - 2x_{35} - 2x_{38} - 2x_{41} - 4x_{60} - 2x_{61} - 4x_{183}x_{196} \\ & + x_{190} \end{aligned}$$

$$\begin{aligned} r_{142} = & 26 + 2x_0^2 + 2x_0x_3 - 2x_0x_{138} - 2x_0x_{147} + x_1^2 - 2x_1x_{174} + x_2^2 - 2x_2x_{165} + x_3^2 - 2x_3x_{147} \\ & + 26x_8^2 - 30x_8x_9 + 40x_8x_{19} + 20x_8x_{22} + 20x_8x_{26} + 20x_8x_{28} + 20x_8x_{30} + 20x_8x_{32} + 40x_8x_{34} \\ & + 20x_8x_{35} + 20x_8x_{38} + 20x_8x_{41} + 40x_8x_{60} + 20x_8x_{61} - 2x_8x_{150} - 10x_8x_{183} + 10x_9^2 - 24x_9x_{19} \\ & - 12x_9x_{22} - 12x_9x_{26} - 12x_9x_{28} - 12x_9x_{30} - 12x_9x_{32} - 24x_9x_{34} - 12x_9x_{35} - 12x_9x_{38} - 12x_9x_{41} \\ & - 24x_9x_{60} - 12x_9x_{61} - 2x_9x_{153} + 6x_9x_{183} + x_{10}^2 - 2x_{10}x_{156} + x_{11}^2 - 2x_{11}x_{159} + x_{12}^2 \\ & - 2x_{12}x_{162} + x_{13}^2 - 2x_{13}x_{168} + x_{14}^2 - 2x_{14}x_{171} + x_{15}^2 - 2x_{15}x_{177} + x_{16}^2 - 2x_{16}x_{180} \\ & + 16x_{19}^2 + 16x_{19}x_{22} + 16x_{19}x_{26} + 16x_{19}x_{28} + 16x_{19}x_{30} + 16x_{19}x_{32} + 32x_{19}x_{34} + 16x_{19}x_{35} \\ & + 16x_{19}x_{38} + 16x_{19}x_{41} + 32x_{19}x_{60} + 16x_{19}x_{61} - 8x_{19}x_{183} + 4x_{22}^2 + 8x_{22}x_{26} + 8x_{22}x_{28} + 8x_{22}x_{30} \\ & + 8x_{22}x_{32} + 16x_{22}x_{34} + 8x_{22}x_{35} + 8x_{22}x_{38} + 8x_{22}x_{41} + 16x_{22}x_{60} + 8x_{22}x_{61} - 4x_{22}x_{183} + 4x_{26}^2 \\ & + 8x_{26}x_{28} + 8x_{26}x_{30} + 8x_{26}x_{32} + 16x_{26}x_{34} + 8x_{26}x_{35} + 8x_{26}x_{38} + 8x_{26}x_{41} + 16x_{26}x_{60} + 8x_{26}x_{61} \\ & - 4x_{26}x_{183} + 4x_{28}^2 + 8x_{28}x_{30} + 8x_{28}x_{32} + 16x_{28}x_{34} + 8x_{28}x_{35} + 8x_{28}x_{38} + 8x_{28}x_{41} + 16x_{28}x_{60} \\ & + 8x_{28}x_{61} - 4x_{28}x_{183} + 4x_{30}^2 + 8x_{30}x_{32} + 16x_{30}x_{34} + 8x_{30}x_{35} + 8x_{30}x_{38} + 8x_{30}x_{41} + 16x_{30}x_{60} \\ & + 8x_{30}x_{61} - 4x_{30}x_{183} + 4x_{32}^2 + 16x_{32}x_{34} + 8x_{32}x_{35} + 8x_{32}x_{38} + 8x_{32}x_{41} + 16x_{32}x_{60} + 8x_{32}x_{61} \\ & - 4x_{32}x_{183} + 16x_{34}^2 + 16x_{34}x_{35} + 16x_{34}x_{38} + 16x_{34}x_{41} + 32x_{34}x_{60} + 16x_{34}x_{61} - 8x_{34}x_{183} + 4x_{35}^2 \\ & + 8x_{35}x_{38} + 8x_{35}x_{41} + 16x_{35}x_{60} + 8x_{35}x_{61} - 4x_{35}x_{183} + 4x_{38}^2 + 8x_{38}x_{41} + 16x_{38}x_{60} + 8x_{38}x_{61} \\ & - 4x_{38}x_{183} + 4x_{41}^2 + 16x_{41}x_{60} + 8x_{41}x_{61} - 4x_{41}x_{183} + 16x_{60}^2 + 16x_{60}x_{61} - 8x_{60}x_{183} + 4x_{61}^2 \\ & - 4x_{61}x_{183} - x_{133}x_{200} - 2x_{137} - 2x_{139} + x_{139}^2 - 2x_{146} - 2x_{148} + x_{148}^2 - 2x_{149} + 2x_{150} - 2x_{151} + x_{151}^2 \end{aligned}$$

$$
\begin{aligned}
&- 2x_{152} - 2x_{154} + x_{154}^2 - 2x_{155} - 2x_{157} + x_{157}^2 - 2x_{158} - 2x_{160} + x_{160}^2 - 2x_{161} - 2x_{163} + x_{163}^2 - 2x_{164} \\
&- 2x_{166} + x_{166}^2 - 2x_{167} - 2x_{169} + x_{169}^2 - 2x_{170} - 2x_{172} + x_{172}^2 - 2x_{173} + x_{175}^2 - 2x_{176} - 2x_{178} \\
&+ x_{178}^2 - 2x_{179} - 2x_{181} + x_{181}^2 - 2x_{182} - 2x_{184} + x_{184}^2
\end{aligned}
$$

$$r_{143} = - x_{95}x_{201} + x_{119}$$

$$r_{144} = - x_{112}x_{202} + x_{119}$$

$$r_{145} = - 1 + x_2 - x_3 - 2x_3x_{210}$$

$$
\begin{aligned}
r_{146} = &- 2 - 25x_8^2 + 30x_8x_9 - 40x_8x_{19} - 20x_8x_{22} - 20x_8x_{26} - 20x_8x_{28} - 20x_8x_{30} - 20x_8x_{32} - 40x_8x_{34} \\
&- 20x_8x_{35} - 20x_8x_{38} - 20x_8x_{41} - 40x_8x_{60} - 20x_8x_{61} - 9x_9^2 + 24x_9x_{19} + 12x_9x_{22} + 12x_9x_{26} \\
&+ 12x_9x_{28} + 12x_9x_{30} + 12x_9x_{32} + 24x_9x_{34} + 12x_9x_{35} + 12x_9x_{38} + 12x_9x_{41} + 24x_9x_{60} + 12x_9x_{61} \\
&- 16x_{19}^2 - 16x_{19}x_{22} - 16x_{19}x_{26} - 16x_{19}x_{28} - 16x_{19}x_{30} - 16x_{19}x_{32} - 32x_{19}x_{34} - 16x_{19}x_{35} \\
&- 16x_{19}x_{38} - 16x_{19}x_{41} - 32x_{19}x_{60} - 16x_{19}x_{61} - 4x_{22}^2 - 8x_{22}x_{26} - 8x_{22}x_{28} - 8x_{22}x_{30} - 8x_{22}x_{32} \\
&- 16x_{22}x_{34} - 8x_{22}x_{35} - 8x_{22}x_{38} - 8x_{22}x_{41} - 16x_{22}x_{60} - 8x_{22}x_{61} - 4x_{26}^2 - 8x_{26}x_{28} - 8x_{26}x_{30} \\
&- 8x_{26}x_{32} - 16x_{26}x_{34} - 8x_{26}x_{35} - 8x_{26}x_{38} - 8x_{26}x_{41} - 16x_{26}x_{60} - 8x_{26}x_{61} - 4x_{28}^2 - 8x_{28}x_{30} \\
&- 8x_{28}x_{32} - 16x_{28}x_{34} - 8x_{28}x_{35} - 8x_{28}x_{38} - 8x_{28}x_{41} - 16x_{28}x_{60} - 8x_{28}x_{61} - 4x_{30}^2 - 8x_{30}x_{32} \\
&- 16x_{30}x_{34} - 8x_{30}x_{35} - 8x_{30}x_{38} - 8x_{30}x_{41} - 16x_{30}x_{60} - 8x_{30}x_{61} - 4x_{32}^2 - 16x_{32}x_{34} - 8x_{32}x_{35} \\
&- 8x_{32}x_{38} - 8x_{32}x_{41} - 16x_{32}x_{60} - 8x_{32}x_{61} - 16x_{34}^2 - 16x_{34}x_{35} - 16x_{34}x_{38} - 16x_{34}x_{41} - 32x_{34}x_{60} \\
&- 16x_{34}x_{61} - 4x_{35}^2 - 8x_{35}x_{38} - 8x_{35}x_{41} - 16x_{35}x_{60} - 8x_{35}x_{61} - 4x_{38}^2 - 8x_{38}x_{41} - 16x_{38}x_{60} \\
&- 8x_{38}x_{61} - 4x_{41}^2 - 16x_{41}x_{60} - 8x_{41}x_{61} - 16x_{60}^2 - 16x_{60}x_{61} - 4x_{61}^2 - x_{138}^2 - x_{147}^2 + 2x_{150} \\
&- x_{150}^2 - x_{153}^2 - x_{156}^2 - x_{159}^2 - x_{162}^2 - x_{165}^2 - x_{168}^2 - x_{171}^2 - x_{174}^2 - x_{177}^2 \\
&- x_{180}^2 + x_{183}^2 - x_{203}^2 - x_{204}^2 - x_{205} - x_{205}^2
\end{aligned}
$$

$$
\begin{aligned}
r_{147} = &- 1 - 2x_{97}^2 - 2x_{101}^2 - 2x_{104}^2 - 2x_{107}^2 - 2x_{110}^2 - 2x_{114}^2 - 2x_{117}^2 + x_{119}^2 - 2x_{121}^2 - 2x_{125}^2 \\
&- 2x_{128}^2 - 2x_{131}^2 - x_{206}^2 - x_{207}^2 - x_{208} - x_{208}^2
\end{aligned}
$$

$$r_{148} = - x_3x_{12} + x_{211}$$

$$
\begin{aligned}
r_{149} = &- 2x_{94}x_{147} + 2x_{99} - 2x_{99}x_{150} - 2x_{102}x_{153} - 2x_{105}x_{156} - 2x_{108}x_{159} - 2x_{111}x_{162} - 2x_{115}x_{168} \\
&- 2x_{118}x_{171} - 2x_{123}x_{177} - 2x_{126}x_{180} - 2x_{129}x_{183} - 2x_{133}x_{213} + 2x_{146}x_{147} + 2x_{147}x_{148} - 2x_{149} \\
&+ 2x_{149}x_{150} + 2x_{150}x_{151} - 2x_{151} + 2x_{152}x_{153} + 2x_{153}x_{154} + 2x_{155}x_{156} + 2x_{156}x_{157} + 2x_{158}x_{159} \\
&+ 2x_{159}x_{160} + 2x_{161}x_{162} + 2x_{162}x_{163} + 2x_{167}x_{168} + 2x_{168}x_{169} + 2x_{170}x_{171} + 2x_{171}x_{172} + 2x_{176}x_{177} \\
&+ 2x_{177}x_{178} + 2x_{179}x_{180} + 2x_{180}x_{181} + 2x_{182}x_{183} + 2x_{183}x_{184} + 4x_{209}x_{213} + x_{212}
\end{aligned}
$$

$$
\begin{aligned}
r_{150} = &- 1 + x_{94}^2 - 2x_{94}x_{146} + x_{99}^2 - 2x_{99}x_{149} + x_{102}^2 - 2x_{102}x_{152} + x_{105}^2 - 2x_{105}x_{155} + x_{108}^2 \\
&- 2x_{108}x_{158} + x_{111}^2 - 2x_{111}x_{161} + x_{115}^2 - 2x_{115}x_{167} + x_{118}^2 - 2x_{118}x_{170} + x_{123}^2 - 2x_{123}x_{176} \\
&+ x_{126}^2 - 2x_{126}x_{179} + x_{129}^2 - 2x_{129}x_{182} + x_{146}^2 - x_{147}^2 + x_{149}^2 + 2x_{150} - x_{150}^2 + x_{152}^2
\end{aligned}
$$

$$
\begin{aligned}
&- x_{153}^2 + x_{155}^2 - x_{156}^2 + x_{158}^2 - x_{159}^2 + x_{161}^2 - x_{162}^2 + x_{167}^2 - x_{168}^2 + x_{170}^2 \\
&- x_{171}^2 + x_{176}^2 - x_{177}^2 + x_{179}^2 - x_{180}^2 + x_{182}^2 - x_{183}^2 - 4x_{209}x_{212} - 2x_{212} + x_{213}
\end{aligned}
$$

$$
\begin{aligned}
r_{151} = \; &- 2x_{95}x_{138} - 2x_{112}x_{165} - 2x_{119}x_{174} - 2x_{133}x_{215} + 2x_{137}x_{138} + 2x_{138}x_{139} + 2x_{164}x_{165} + 2x_{165}x_{166} \\
&+ 2x_{173}x_{174} + 2x_{174} + 2x_{174}x_{175} + 4x_{209}x_{215} + x_{214} - x_{215}
\end{aligned}
$$

$$
\begin{aligned}
r_{152} = \; & x_{95}^2 - 2x_{95}x_{137} + x_{112}^2 - 2x_{112}x_{164} + x_{119}^2 - 2x_{119}x_{173} + x_{137}^2 - x_{138}^2 + x_{164}^2 - x_{165}^2 \\
&+ x_{173}^2 - x_{174}^2 - 4x_{209}x_{214} - x_{214} + x_{215}
\end{aligned}
$$

C Complete monic-elimination tables

For each profile the listed pair (r, v) means solve original row r_r for original coordinate x_v. All coefficients used for solving are ± 1. Use recursive acyclic substitution, not a change in quantifier domain. Coordinates outside (12) and outside the listed pivots remain free; append one new integer per block part. These are fixed lists, with no data-dependent iteration in the definition of the final polynomial.

C.1 143 integers, degree 8

Block partition: (0; 1; 2; 3; 4). There are 68 deleted monic coordinates and 80 final residuals.

(71, 86)	(50, 65)	(79, 98)	(37, 52)	(64, 79)	(73, 88)
(57, 72)	(53, 68)	(81, 101)	(83, 107)	(61, 75)	(52, 67)
(84, 110)	(89, 125)	(54, 69)	(88, 121)	(18, 33)	(85, 114)
(78, 92)	(41, 56)	(136, 199)	(30, 45)	(91, 131)	(90, 128)
(51, 66)	(82, 104)	(86, 117)	(59, 74)	(55, 70)	(43, 58)
(49, 64)	(35, 50)	(93, 136)	(39, 54)	(62, 76)	(14, 29)
(76, 91)	(16, 31)	(32, 47)	(140, 189)	(134, 198)	(25, 40)
(0, 7)	(132, 197)	(10, 25)	(22, 37)	(27, 42)	(28, 43)
(6, 21)	(115, 166)	(111, 160)	(123, 178)	(107, 154)	(119, 172)
(125, 181)	(95, 139)	(105, 151)	(103, 148)	(121, 175)	(109, 157)
(117, 169)	(113, 163)	(127, 184)	(1, 1)	(3, 16)	(20, 35)
(13, 28)	(143, 119)				

C.2 119 integers, degree 16

Block partition: (0, 1, 2; 3, 4). There are 89 deleted monic coordinates and 56 final residuals.

(53, 68)	(138, 191)	(52, 67)	(64, 79)	(54, 69)	(0, 7)
(136, 199)	(50, 65)	(62, 76)	(72, 87)	(49, 64)	(63, 78)
(55, 70)	(67, 82)	(85, 114)	(90, 128)	(81, 101)	(82, 104)
(89, 125)	(80, 97)	(86, 117)	(88, 121)	(84, 110)	(83, 107)
(91, 131)	(10, 21)	(48, 63)	(29, 44)	(76, 91)	(78, 92)
(74, 89)	(16, 31)	(132, 197)	(37, 52)	(32, 47)	(44, 59)
(27, 42)	(93, 136)	(61, 75)	(117, 169)	(105, 151)	(119, 172)
(109, 157)	(127, 184)	(107, 154)	(111, 160)	(125, 181)	(1, 1)
(113, 163)	(103, 148)	(115, 166)	(123, 178)	(41, 56)	(18, 33)
(14, 29)	(51, 66)	(46, 61)	(20, 35)	(23, 38)	(13, 28)
(56, 71)	(3, 16)	(25, 40)	(121, 175)	(95, 139)	(39, 54)
(71, 86)	(141, 190)	(87, 122)	(79, 98)	(92, 135)	(69, 84)
(143, 119)	(65, 80)	(59, 74)	(134, 198)	(75, 90)	(28, 43)
(43, 58)	(34, 49)	(31, 46)	(24, 39)	(36, 51)	(21, 36)
(5, 20)	(58, 73)	(140, 189)	(148, 211)	(9, 24)	

C.3 112 integers, degree 24

Block partition: $(0, 1, 2, 3, 4)$. There are 95 deleted monic coordinates and 49 final residuals.

$(53, 68)$	$(138, 191)$	$(52, 67)$	$(64, 79)$	$(54, 69)$	$(0, 7)$
$(136, 199)$	$(50, 65)$	$(62, 76)$	$(72, 87)$	$(49, 64)$	$(63, 78)$
$(55, 70)$	$(67, 82)$	$(85, 114)$	$(90, 128)$	$(81, 101)$	$(82, 104)$
$(89, 125)$	$(80, 97)$	$(86, 117)$	$(88, 121)$	$(84, 110)$	$(83, 107)$
$(91, 131)$	$(10, 21)$	$(48, 63)$	$(29, 44)$	$(76, 91)$	$(78, 92)$
$(74, 89)$	$(16, 31)$	$(132, 197)$	$(37, 52)$	$(32, 47)$	$(44, 59)$
$(27, 42)$	$(93, 136)$	$(61, 75)$	$(41, 56)$	$(18, 33)$	$(14, 29)$
$(51, 66)$	$(56, 71)$	$(69, 84)$	$(25, 40)$	$(59, 74)$	$(39, 54)$
$(71, 86)$	$(141, 190)$	$(1, 1)$	$(3, 16)$	$(20, 35)$	$(13, 28)$
$(23, 38)$	$(46, 61)$	$(87, 122)$	$(79, 98)$	$(92, 135)$	$(117, 169)$
$(105, 151)$	$(121, 175)$	$(119, 172)$	$(109, 157)$	$(127, 184)$	$(107, 154)$
$(111, 160)$	$(95, 139)$	$(125, 181)$	$(113, 163)$	$(103, 148)$	$(115, 166)$
$(123, 178)$	$(143, 119)$	$(65, 80)$	$(134, 198)$	$(75, 90)$	$(28, 43)$
$(43, 58)$	$(34, 49)$	$(31, 46)$	$(24, 39)$	$(36, 51)$	$(21, 36)$
$(5, 20)$	$(58, 73)$	$(140, 189)$	$(11, 26)$	$(57, 72)$	$(148, 211)$
$(145, 2)$	$(68, 83)$	$(60, 77)$	$(9, 24)$	$(137, 188)$	

C.4 107 integers, degree 32

Block partition: $(0, 1, 2, 3, 4)$. There are 100 deleted monic coordinates and 44 final residuals.

$(53, 68)$	$(138, 191)$	$(52, 67)$	$(64, 79)$	$(54, 69)$	$(0, 7)$
$(136, 199)$	$(50, 65)$	$(62, 76)$	$(72, 87)$	$(49, 64)$	$(63, 78)$
$(55, 70)$	$(67, 82)$	$(85, 114)$	$(90, 128)$	$(81, 101)$	$(82, 104)$
$(89, 125)$	$(80, 97)$	$(86, 117)$	$(88, 121)$	$(84, 110)$	$(83, 107)$
$(91, 131)$	$(10, 21)$	$(48, 63)$	$(29, 44)$	$(76, 91)$	$(78, 92)$
$(74, 89)$	$(16, 31)$	$(132, 197)$	$(37, 52)$	$(32, 47)$	$(44, 59)$
$(27, 42)$	$(93, 136)$	$(61, 75)$	$(41, 56)$	$(18, 33)$	$(14, 29)$
$(51, 66)$	$(56, 71)$	$(69, 84)$	$(25, 40)$	$(59, 74)$	$(39, 54)$
$(71, 86)$	$(141, 190)$	$(1, 1)$	$(3, 16)$	$(20, 35)$	$(13, 28)$
$(23, 38)$	$(46, 61)$	$(87, 122)$	$(79, 98)$	$(92, 135)$	$(117, 169)$
$(105, 151)$	$(121, 175)$	$(119, 172)$	$(109, 157)$	$(127, 184)$	$(107, 154)$
$(111, 160)$	$(95, 139)$	$(125, 181)$	$(113, 163)$	$(103, 148)$	$(115, 166)$
$(123, 178)$	$(143, 119)$	$(65, 80)$	$(134, 198)$	$(75, 90)$	$(28, 43)$
$(43, 58)$	$(34, 49)$	$(31, 46)$	$(24, 39)$	$(36, 51)$	$(131, 0)$
$(21, 36)$	$(5, 20)$	$(42, 57)$	$(58, 73)$	$(140, 189)$	$(11, 26)$
$(57, 72)$	$(148, 211)$	$(145, 2)$	$(68, 83)$	$(60, 77)$	$(9, 24)$
$(137, 188)$	$(47, 62)$	$(8, 23)$	$(4, 19)$		

C.5 102 integers, degree 64

Block partition: $(0, 1, 2, 3, 4)$. There are 105 deleted monic coordinates and 39 final residuals.

$(72, 87)$	$(70, 85)$	$(74, 89)$	$(68, 83)$	$(136, 199)$	$(65, 80)$
$(79, 98)$	$(138, 191)$	$(62, 76)$	$(48, 63)$	$(0, 7)$	$(51, 66)$
$(76, 91)$	$(78, 92)$	$(58, 73)$	$(54, 69)$	$(50, 65)$	$(53, 68)$
$(61, 75)$	$(63, 78)$	$(55, 70)$	$(52, 67)$	$(49, 64)$	$(129, 132)$
$(132, 197)$	$(84, 110)$	$(90, 128)$	$(81, 101)$	$(82, 104)$	$(89, 125)$
$(91, 131)$	$(85, 114)$	$(83, 107)$	$(88, 121)$	$(86, 117)$	$(36, 51)$
$(140, 189)$	$(141, 190)$	$(31, 46)$	$(34, 49)$	$(38, 53)$	$(29, 44)$
$(40, 55)$	$(56, 71)$	$(6, 21)$	$(22, 37)$	$(44, 59)$	$(25, 40)$
$(16, 31)$	$(14, 29)$	$(10, 25)$	$(18, 33)$	$(80, 97)$	$(87, 122)$
$(148, 211)$	$(67, 82)$	$(64, 79)$	$(59, 74)$	$(75, 90)$	$(93, 136)$
$(71, 86)$	$(134, 198)$	$(115, 166)$	$(103, 148)$	$(117, 169)$	$(113, 163)$
$(123, 178)$	$(95, 139)$	$(107, 154)$	$(119, 172)$	$(125, 181)$	$(109, 157)$
$(111, 160)$	$(127, 184)$	$(121, 175)$	$(105, 151)$	$(24, 39)$	$(42, 57)$
$(43, 58)$	$(28, 43)$	$(27, 42)$	$(32, 47)$	$(57, 72)$	$(92, 135)$
$(60, 77)$	$(66, 81)$	$(21, 36)$	$(9, 24)$	$(5, 20)$	$(137, 188)$
$(8, 23)$	$(1, 1)$	$(3, 16)$	$(46, 61)$	$(20, 35)$	$(143, 119)$
$(13, 28)$	$(131, 0)$	$(37, 52)$	$(47, 62)$	$(145, 2)$	$(130, 185)$
$(41, 56)$	$(139, 133)$	$(4, 19)$			

References

[1] ProveIt research package 172, *Pooled powers and quadratic projections for ZFC consistency*, 25 October 2026. The signed arithmetic component, its ordinary proof, source calibration and affine216 chart are retained in this release. https://github.com/VladimirReshetnikov/ProveIt/tree/ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/172-pooled-powers-round21

[2] ProveIt research package 168, *Reciprocal pooling of distinct-base Pell certificates*, 25 October 2026, especially `proofs/RECIPROCAL_POOLING.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/168-reciprocal-pooling

[3] ProveIt research package 158, *Pooling exact power certificates for Diophantine consistency*, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers

[4] ProveIt research package 152, *Lucas corrections and elimination of arithmetic witnesses*, 25 October 2026, higher-degree component. https://github.com/VladimirReshetnikov/ProveIt/tree/ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/152-lucas-corrections

[5] Vladimir Reshetnikov, *ProveIt: Con(ZFC) in Diophantine form*, repository corpus and comparison README, inspected at the pinned snapshot stated in Section 1. https://github.com/VladimirReshetnikov/ProveIt/tree/ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b/Computability/DiophantineConsistency/con-zfc-diophantine-forms